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PrevenTech™ Telematics System Installation Manual

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System and Hardware Overview

This instruction manual details the installation of the PrevenTech™ telematics system. While efforts have been made to make this manual as comprehensive as possible, installations must be tailored to the particular application/equipment model.

The PrevenTech™ Kit contains the following hardware:

- a. PrevenTech™ telematics box
- b. Cellular/GPS/Bluetooth antenna
- c. Main wiring harness
- d. Antenna mounting hardware

The PrevenTech™ kit does not contain all the hardware needed for the installation. SIM cards and connection wiring harnesses **must** be purchased separately.



Figure 1, PrevenTech™ Kit Hardware.

Environmental and Physical Requirements Operation and Storage Requirements:

Table 1, Operation and Storage Requirements

Circuitry Operation Temperature Range	-5°C to +70°C [23 °F to 158 °F]
Storage Temperature Range	-30°C to +85°C [86 °F to 185 °F]
Humidity	Up to 95 % - non-condensing

Pre-Installation Steps

Prior to taking the PrevenTech™ hardware to the customer site, a bench test must be performed to verify that the system is functional and the procured SIM card is compatible with the system.

1. Register box by contacting CPS team at connectedsol.support@cummins.com. On the PrevenTech™ box, provide ID and IMEI numbers. Also, needed are the engine serial number (ESN), ECM calibration, hp rating, equipment OEM/model, unit number, location/mine site, and SIM card information.



Figure 2, Identification label on Bottom of PrevenTech™ Box.

2. Remove telematics box cover and verify that dip switches are switched off (towards outboard position or to the “right” if the harness connector is at the top). These switches are internal resistors that are **not** needed.

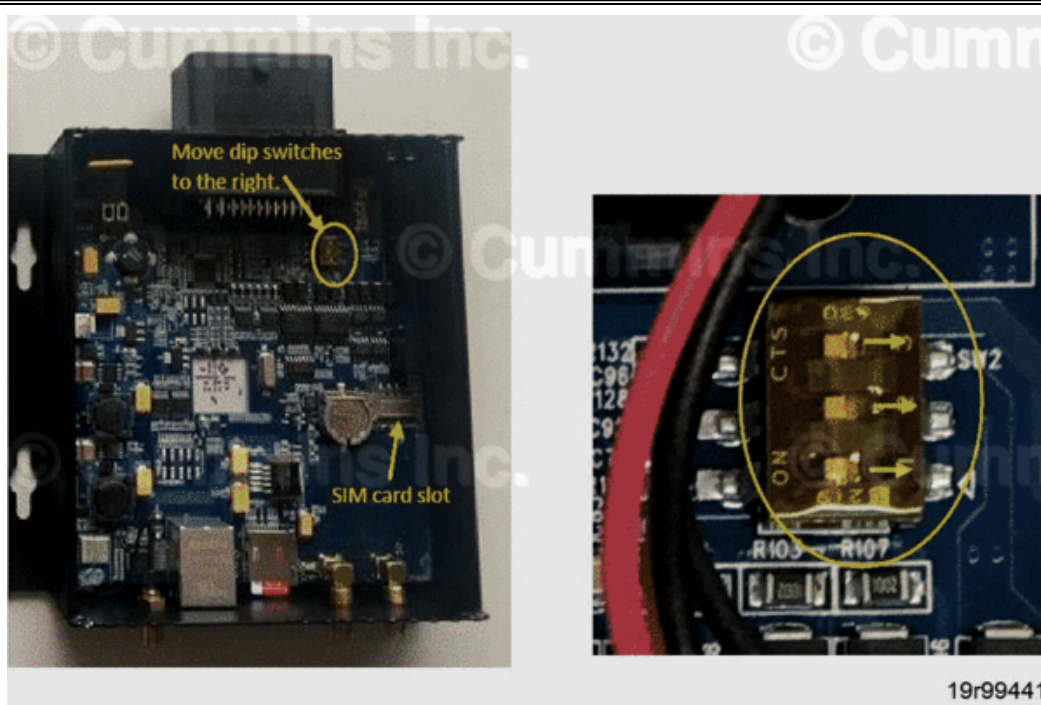


Figure 3, Interior of PrevenTech™ Box. Dip Switch Location Highlighted.

3. Procure 3G GSM SIM card (Standard 2FF size) or backward compatible 4G LTE SIM card. Note that the box will **not** work using a 4G card or a card that is configured to the CDMA network. Verify that there is adequate 3G coverage in the area. When activating the card, the IMEI number on the box label is required. Recommend that the customer and equipment is registered and access to www.preventech.cummins.com is activated. If the customer has **not** ordered SIM cards with active data plan, use a compatible SIM card with activated data plan for operational validation.
4. Install SIM card with notched tab on the outer side



Figure 4, Interior of PrevenTech™ Box With SIM Card Installed.



Figure 5, Clock Battery Location

5. Connect Antenna to Telematics box and place in an elevated position to get cellular signal.
6. Connect wiring harness to box. Connect VINPOT lead to 12/24V power source and GND lead to negative connection. Apply power. After 5 minutes, the Green Light should be Steady On. The Red Light should be Flashing (2 seconds OFF – 0 to 5 seconds ON). If the Red light is Steady On, it means that there is No cell connectivity. If this occurs, reference Appendix A for troubleshooting.
7. Determine J1939 CAN Bus connection strategy. There are a couple different methods to connect to the engine J1939 CAN Bus. This strategy chosen will determine what Peripheral Hardware needs to be ordered.
 - Option 1: Node off OEM J1939 CAN Bus

- This is the method most commonly used. However, there have been instances of OEM's disconnecting the nodes during maintenance activities and **not** reconnecting them. **Must** order a Gender Changer and Y Connector.

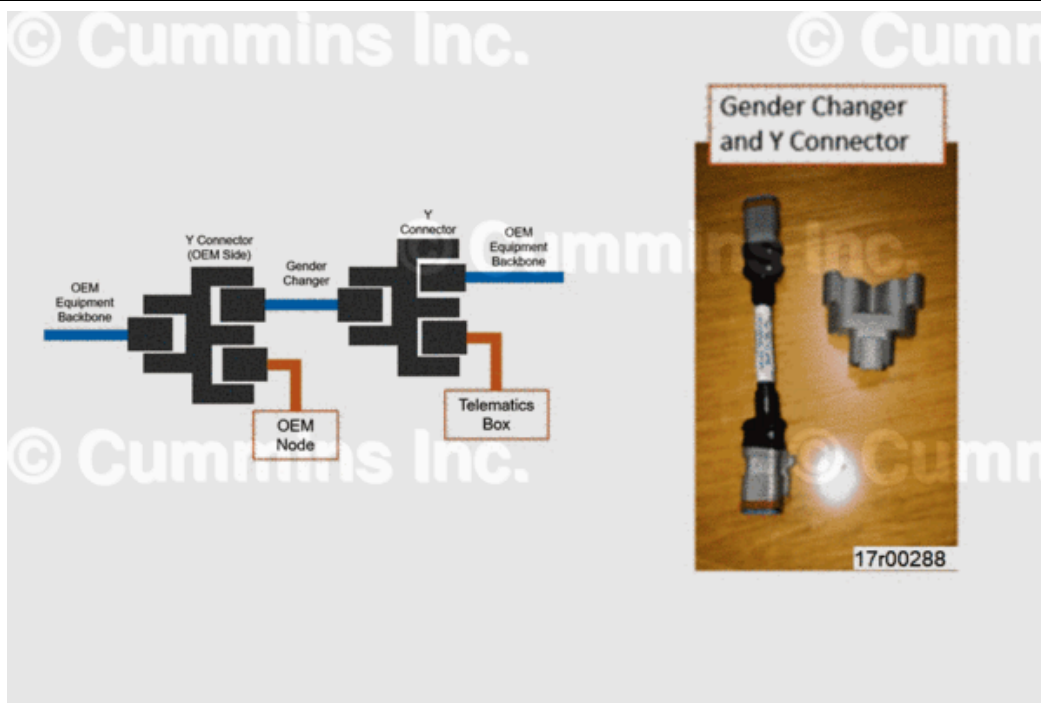


Figure 6, CAN Bus Node

- Option 2: Extend Engine Backbone to Telematics Box Harness.
 - This requires additional wiring harness routing. At present there is a 3 meter and 6 meter harness released and available. Order required extension harness, Y connector, and 120 Ohm terminating resistor.

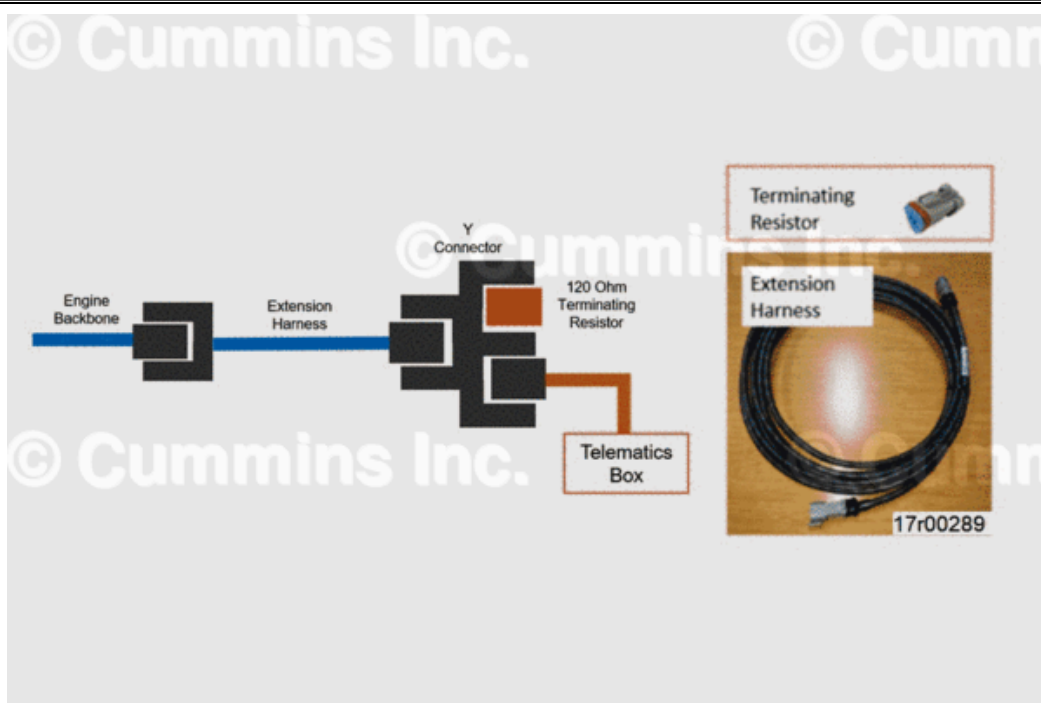


Figure 7, Extension Wiring Harness

8. General installation hardware is not included in the kit. Items required to perform complete installation that are not included in the kit. These items will vary depending on application/model type.

- Zip ties
- Fuse holders
- 5 amp fuse
- Various electrical terminals/butt splices
- Gland Nuts
- Additional 18AWG Wire (If extension of wiring harness leads needed)

Installation Procedures

1. Select J1939 Public CAN Bus 3-Pin Connection

- It is possible for engines and equipment to have multiple CAN networks within their architecture – Public vs. Private. Identifying the Public J1939 CAN Bus is essential prior to installation of the PrevenTech™ telematics box. To verify the correct CAN Bus has been selected, contact the equipment OEM. Alternatively, check continuity between the selected 3-pin connector and the 9-Pin service connector.
 - Continuity between Pin A on 3-Pin Connector and Pin C on 9-Pin Service Connector
 - Continuity between Pin B on 3-Pin Connector and Pin D on 9-Pin Service Connector

2. Verify J1939 CAN Bus resistance by measuring between Positive and Negative terminal on the 9-Pin service connector (Pins C and D in picture below). If resistance measurement is 120 ohms, an additional terminating resistor **must** be added to the backbone. If the resistance measurement is less than 60 ohms, then a resistor **must** be removed.

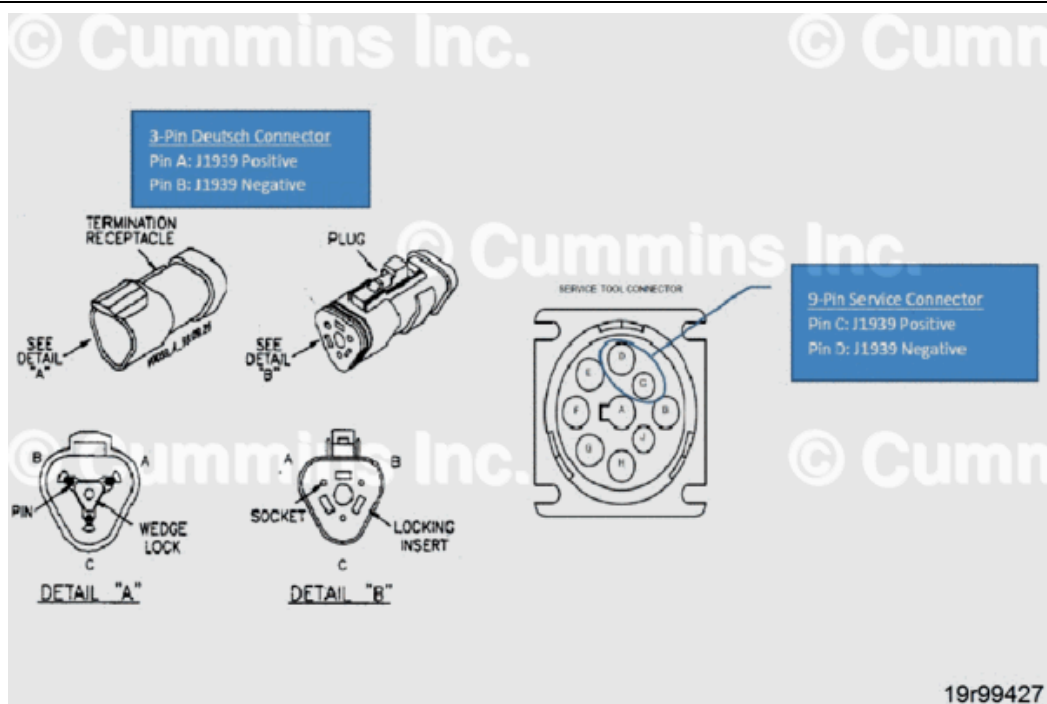


Figure 8, Connector Pin Diagram

Install PrevenTech™ Telematics Box

- Installation location must meet the following environment criteria:
 - **Not** exposed to liquids of any kind
 - **Not** exposed to direct sunlight
 - **Not** attached to structures with excessive vibration
 - Away from excessive heat sources
 - Away from sources of excessive cold such as air conditioning systems
 - Dust free environment, if possible
 - Consider proximity to Power/Ground/Ignition sources

- Mount telematics box using mounting hardware.

4. Connect Wiring Harness

- The PrevenTech™ wiring harness has several wires that are not utilized in the existing configuration. Make the following connections:
 - VINPUT (Red) – Connect to 12/24V power source
 - GND (Black) – Connect to negative power terminal
 - IGN_14 (Yellow) – Connect to Ignition power source
 - DI_VDD (Green) – Install fuse holder and 5 Amp fuse in line and connect to 12/24V power source
 - This lead might not be present on all wiring harnesses
 - DI_GND (Pink) – Connect to negative power terminal
 - This lead might not be present on all wiring harnesses
 - CHASSIS_GND (Black) – Connect to common ground

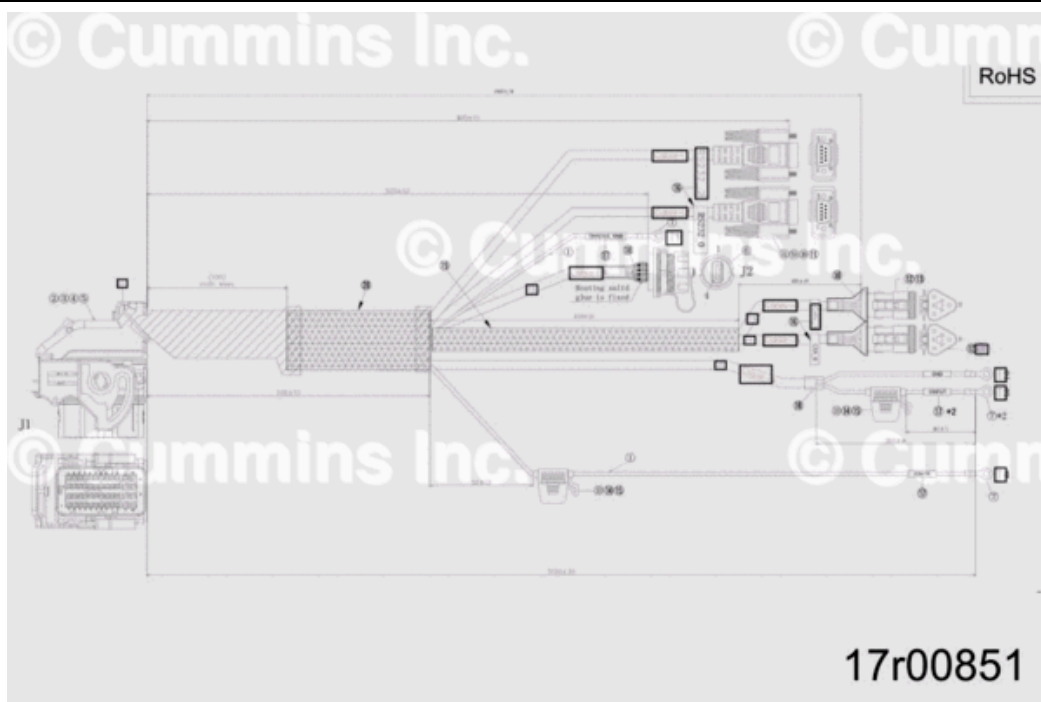


Figure 9, Wiring Harness Breakout

Note : Remove pin K4 on telematics box using a long nose small plier and slowly with a firm grip moving it in an up and down direction until it breaks. This prevents communication errors due to an internal harness miswiring issue of DO (Digital Outputs) wires.

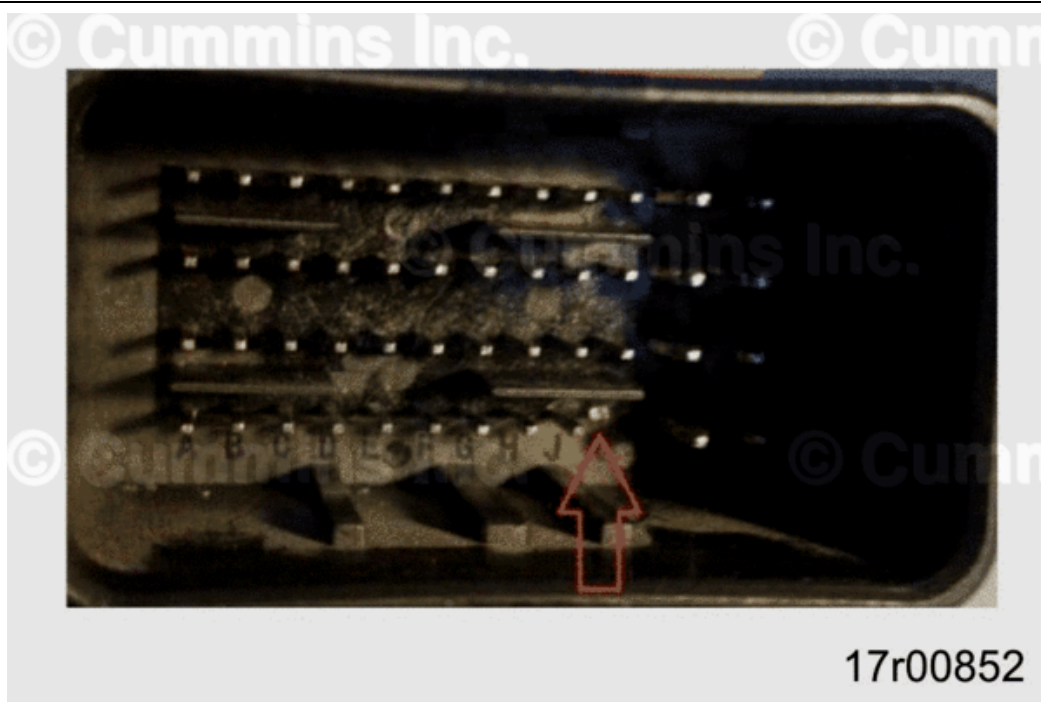


Figure 10, Telematics Box 48 Pin Connector Layout

- Connect wiring harness to PrevenTech™ Telematics Box by inserting connector and then rotating lever into locked position.

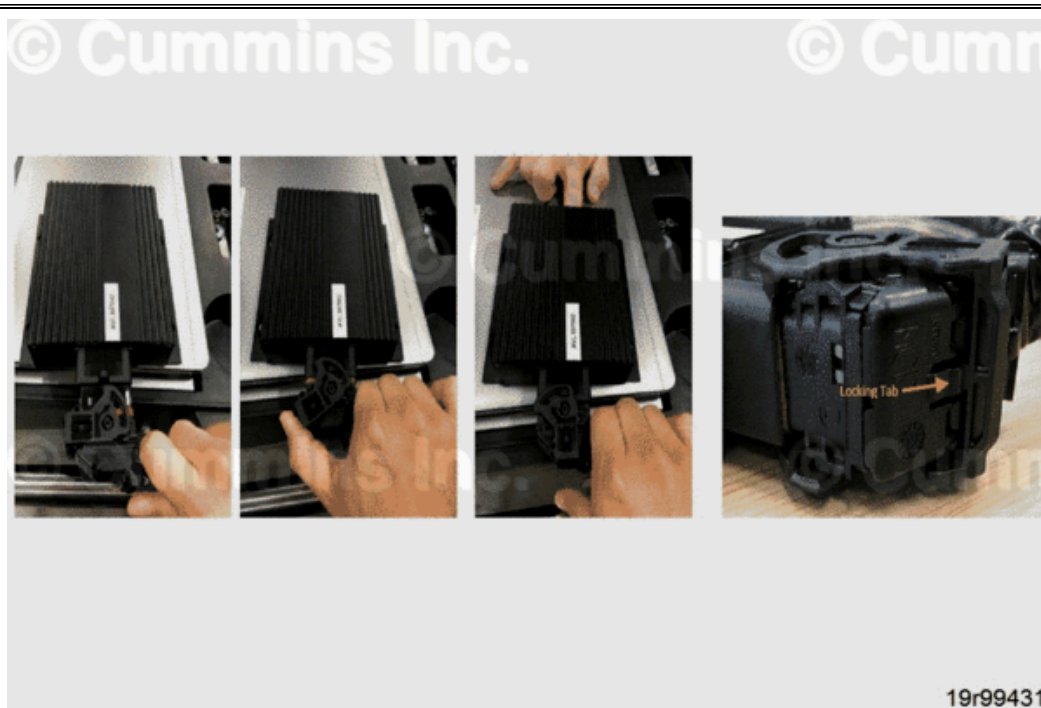


Figure 11, PrevenTech™ Wiring Harness Connection to PrevenTech™ Telematics Box

5. Install Antenna

- GSM / GPS / WIFI antenna must be installed on the outside of the equipment, with direct line of sight to the sky.
- Installed at least 40 cm from equipment operator
- Consider wiring harness routing strategy and proximity to PrevenTech™ Telematics Box
- Route antenna connectors through gasket, bracket, and retaining nut.

- Secure to bracket and mount using two U-bolts provided.
- Route antenna connectors to PrevenTech™ telematics box and match the connectors to the correct port on the box. Note: Antenna connector labeled “WIFI” **must** be connected to telematics box connector labeled “BT”.

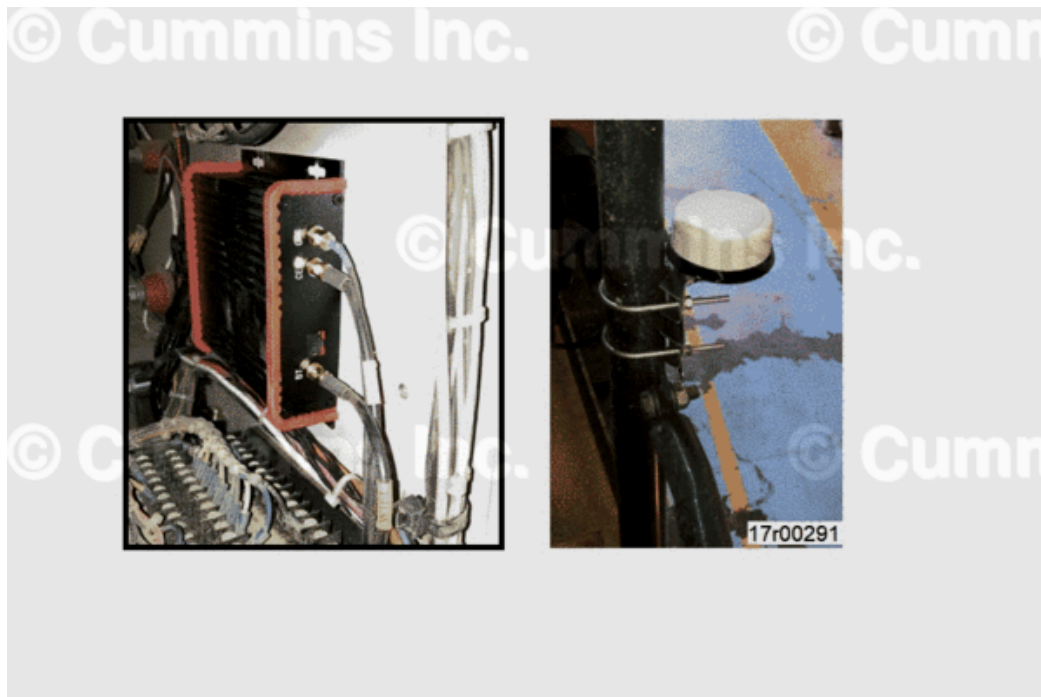


Figure 12, Antenna Harness Connected to Telematics Box and Antenna Mount on Exterior Of Application.

- Create a strain relief by allowing at least 1 foot of cabling between the antennae ports and the placement of the strain relief.
- Secure wiring harness along route using P-Clips or zip ties. Excess wiring harness should be neatly coiled up and secured to prevent from interfering with other components.

6. Apply power and after 5 mins verify that LED lights are on in accordance with the following tables. If LEDs do not give proper indication, reference Appendix A for troubleshooting steps.

- Will turn on after approximately 30 seconds as the power has been supplied. If the equipment is receiving frames through the CAN0 port, the LED will flash at high speed indicating transmission.

Green LED Indicator – CAN Operation

State	Indication
Steady on	Operating system initialized
Flashing	Data is being transmitted on CAN0
Off	Operating system not initialized

- The red LED is used to indicate the communication status of the device. During the first 5 minutes since the device has been powered, the LED will be off. If the telematics box has not achieved connectivity after 5 minutes, the LED will light up steadily. In contrast, if 3G / GSM connectivity is stable for 5 minutes, the LED will flash every 2 seconds.

Red LED Indicator – Network Connection

State	Indication
Steady on (after 5 minutes)	No 3G/GSM connectivity
Flashing (2 seconds OFF – 0 to 5 seconds ON)	Stable 3G/GSM connectivity

Off	Only first 5 minutes after powered on
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7. After equipment registration is verified, use the PrevenTech dashboard to verify GPS location and communication is active. If communication cannot be established, perform troubleshooting steps or contact Cummins CARE.

APPENDIX A: Troubleshooting		
STEP 1A. Check for faulty equipment visibility on PrevenTech™ dashboard		
Conditions: While equipment is on		
Action	Specification/Repair	Next Step
Log on to PrevenTech™ dashboard, https://preventech.cummins.com ; Check dashboard for operation	Is equipment in question visible on dashboard? Yes	2A
	Is equipment in question visible on dashboard? No	1B
STEP 1B. Check for visibility of other equipment on PrevenTech™ dashboard		
Conditions: While equipment is On		
Action	Specification/Repair	Next Step
Check dashboard for visibility of other equipment	Is other equipment visible on dashboard? Yes Note: Not an issue with dashboard	2A
	Is other equipment visible on dashboard? No	Contact: PrevenTech™ Regional Support Contact or care.cummins.com
STEP 2. Identify state of LEDs		
STEP 2A. Green LED flashing		

Conditions: While equipment is on

Note:

30 seconds after power is supplied; PrevenTech™ telematics box green LED will start to flash at high speed when receiving frames through PrevenTech™ telematics J1939 connection

Action	Specification/Repair	Next Step
Inspect green LED	Is green LED flashing? Yes Note: Data is being transmitted on PrevenTech™ telematics box J1939 connection; PrevenTech™ telematics box should be connected to engine's J1939 connection, reference step 5A	2C
	Is green LED flashing? No	2B

STEP 2B. Green LED steady on

Conditions: While equipment is on

Action	Specification/Repair	Next Step
Inspect green LED	Is green LED steady on? Yes Note: Operating System initialized	5A
	Is green LED steady on? No Note: Operating System not initialized	3A

STEP 2C. Red LED flashing

Conditions: While equipment is on

Action	Specification/Repair	Next Step
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Inspect red LED	Is red LED flashing? Yes Note: Stable 3G/GSM connectivity	Check dashboard configuration information Contact: PrevenTech™ Regional Support Contact or care.cummins.com
	Is red LED flashing? No	2D

STEP 2D. Red LED steady on

Conditions: While equipment is on Note: Red LED will be off for the first 5 minutes after power has been supplied to PrevenTech™ telematics box		
Action	Specification/Repair	Next Step
Inspect red LED	Is red LED steady on? Yes Note: No 3G/GSM connectivity	6A
	Is red LED steady on? No	3A

STEP 3. Identify state of power supply

Note : PrevenTech™ telematics box internal cartridge battery will hold charge for period time after external power has been removed and will function as intended; when troubleshooting the external power supply, disconnect PrevenTech™ telematics box internal cartridge battery. If needed refer to Step 4 for PrevenTech™ telematics box internal battery fault finding, i.e. data time stamp variation on PrevenTech™ dashboard.

STEP 3A. Fuse condition

Conditions: While equipment is off		
Action	Specification/Repair	Next Step
Inspect inline fuse	Is fuse open? Yes Repair: Replace fuse	Repair Complete
	Is fuse open? No	3B

STEP 3B. Power supply, ignition, and chassis ground wire condition

Conditions: While equipment is off

Action	Specification/Repair	Next Step
Carefully inspect main harness wiring condition Reference Appendix H above.	Are power supply, ignition, and chassis ground wires in good condition with no damage, i.e. no tears? Yes	3C
	Are power supply, ignition, and chassis ground wires in good condition with no damage? No Repair: Replace PrevenTech™ telematics box main harness	Repair Complete

STEP 3C. Power supply, ignition, and chassis ground connections

Conditions: While equipment is off

Action	Specification/Repair	Next Step
Carefully inspect main harness connections	Are power supply, ignition, and chassis ground wires connected? Yes	3D
	Are power supply, ignition, and chassis ground wires connected? No Repair: Connect wire(s) that are not connected	Repair Complete

STEP 3D. Main harness connection to PrevenTech™ telematics box

Conditions: While equipment is off

Action	Specification/Repair	Next Step
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Inspect main harness connection to PrevenTech™ telematics box. Reference Step 5 in Installation Instructions: PrevenTech™ Telematics Hardware above.	Is the main harness connected securely to PrevenTech™ telematics box? Yes	3E
	Is the main harness connected securely to PrevenTech™ telematics box? No Repair: Connect and securely lock main harness	Repair Complete

STEP 3E. Condition of power supplied to motherboard

Conditions: While power is supplied to PrevenTech™ telematics box		
Action	Specification/Repair	Next Step
Open PrevenTech™ box and inspect LAN port on motherboard, reference Figure 39 above	Is LAN port light on? Yes	3F
	Is LAN port light on? No Repair: Replace PrevenTech™ telematics box	Repair Complete

STEP 3F. Condition of LED(s) internal wiring

Conditions: While power is supplied to PrevenTech™ telematics box		
Action	Specification/Repair	Next Step
Inspect LED(s) internal wiring, reference Figure 39 above	Are LED(s) on? No Repair: Replace PrevenTech™ telematics box	Repair complete

STEP 4. PrevenTech™ internal battery conditions

STEP 4A. Internal clock (round) battery condition

Conditions: While equipment is off

Action	Specification/Repair	Next Step
Check dashboard data time stamps Reference Figure 5 above.	Is there ambiguity in the data time stamp/real time clock? Yes Note: Remove round battery using a thin flathead screwdriver and replace, check PrevenTech™ dashboard for functionality	Repair Complete

STEP 5. Equipment J1939 connections

STEP 5A. Inspect equipment and PrevenTech™ telematics box J1939 connection

Conditions: While equipment is off

Action	Specification/Repair	Next Step
Inspect PrevenTech™ Telematics box J1939 connection with equipment	Is PrevenTech™ telematics box J1939 plug connected to equipment's J1939 public data connection? Yes	5C
	Is PrevenTech™ telematics box J1939 plug connected to equipment's J1939 public data connection? No	5B

STEP 5B. Equipment J1939 Public Connection

Conditions: While equipment is off

Action	Specification/Repair	Next Step
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Measure continuity from 9-pin service connector to 3-pin Deutsch connector	Was continuity confirmed? Yes Repair: Connect PrevenTech™ telematics box J1939 connection to public data J1939 connection on equipment	Repair Complete
	Was continuity confirmed? No Note: Repeat this step until equipment J1939 public connection is located	Repeat Step

STEP 5C. Measure equipment resistance

Conditions: While equipment is off		
Action	Specification/Repair	Next Step
Disconnect the J1939 connection; using the equipment J1939 or 9 Pin service connector measure the equipment resistance, reference Figure 1 above.	Is the measured resistance 60 Ohms? Yes	5E
	Is the measured resistance 60 Ohms? No	5D

STEP 5D. Inspect J1939 backbone terminal resistors

Conditions: While equipment is off		
Action	Specification/Repair	Next Step

Inspect J1939 connection terminal resistors; measure resistance. Reference Figure 7 and Figure 26 above. Note: ONLY two 120 Ohms resistors needed in parallel on backbone to achieve a total resistance of 60 Ohms	Is the measured resistance in each resistor 120 Ohms? Yes	5E
	Is the measured resistance in each resistor 120 Ohms? No Repair: Discard and replace resistor(s)	Repair Complete

STEP 5E. Measure PrevenTech™ telematics box resistance

Conditions: While equipment is off		
Action	Specification/Repair	Next Step
Measure resistance on PrevenTech™ J1939 connection, reference Figure 1 above.	Is the measured resistance approximately 50 K-Ohms? No	5F
	Is the measured resistance approximately 50 K-Ohms? Yes Note: No PrevenTech™ telematics resistance issue	Repair Complete

STEP 5F. PrevenTech™ telematics box resistance correction

Conditions: While equipment is off		
Action	Specification/Repair	Next Step

Measure resistance on PrevenTech™ 1939 connection, reference Figure 1 above.	Is the measured resistance approximately 0 Ohms? Yes Note: Short in box Repair: Replace PrevenTech™ telematics box	Repair Complete
	Is the measured resistance approximately 120 Ohms? Yes Repair: Ensure internal resistance dip switches are all to the right of the battery. Reference Step 13 in Installation Instructions: PrevenTech™ Telematics Hardware above.	Repair Complete

STEP 6. Data transmission to PrevenTech™ dashboard

STEP 6A. Antennae connections

Conditions: While equipment is off		
Action	Specification/Repair	Next Step
Inspect antennae connections on PrevenTech™ telematics box, reference Step 11 in Installation Instructions: PrevenTech™ Telematics Hardware above	Are the antennae wires connect securely and correct to PrevenTech™ telematics box? Yes	6B
	Are the antennae wires connect securely and correct to PrevenTech™ telematics box? No Repair: Correctly wire the antennae wires to the correct ports and securely connect.	Repair Complete

STEP 6B. Condition of antennae wires

Conditions: While equipment is off

Action	Specification/Repair	Next Step
Inspect wires between PrevenTech™ telematics box and antennae round transmitter, reference Step 11 in Installation Instructions: PrevenTech™ Telematics Hardware above	Are one or more antennae wires damaged, i.e. abrasion to conduit, round transmitter physical damage? Yes Repair: Replace and route antennae to PrevenTech™ Telematics box	Repair Complete
	Are one or more antennae wires damaged, i.e. abrasion to conduit, round transmitter physical damage? No	6C

STEP 6C. SIM card status

Conditions: While equipment is off

Action	Specification/Repair	Next Step
Remove SIM card Reference Figure 14 above.	Are SIM card and PrevenTech™ telematics box motherboard contacts dirty or dull? Yes Repair: Clean all contacts, reinstall SIM, and check for data transmission	Repair Complete
	Are SIM card and PrevenTech™ telematics box motherboard contacts dirty or dull? No	6D

STEP 6D. SIM card data

Conditions: While equipment is off

Note: Consider using a mobile phone compatible with SIM card, same network provider, to check data functionality; if data transmission is established, replacement of PrevenTech™ telematics box may be necessary

Action	Specification/Repair	Next Step
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Check data usage with network provider	Has data plan maximum limit been reached? Yes Repair May consider upgrading to a higher data plan	Repair Complete
	Has data plan maximum limit been reached? No Repair: Corrupt SIM card; obtain another SIM card and re-configure PrevenTech™ Telematics box	Repair Complete

Appendix B: Installation Examples

The following pictures document some of the installations conducted on various applications. These are for reference only. Each installation requires evaluation to determine what the correct strategy is for that particular unit.

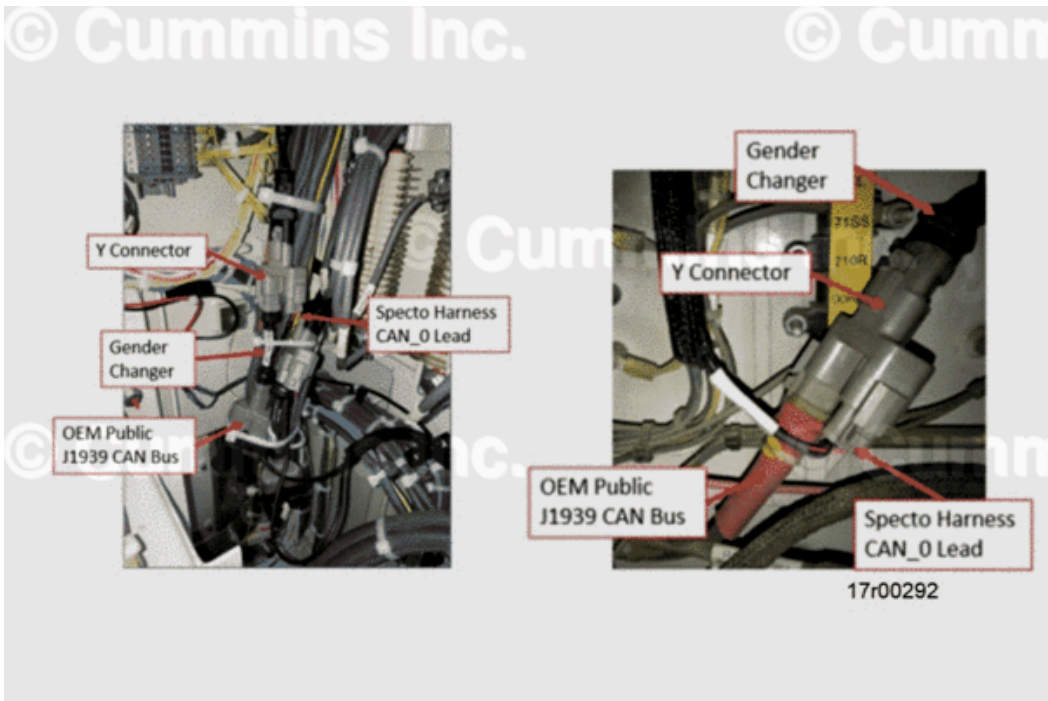


Figure 13, Connecting into J1939 CAN Bus using Gender Changer and Y Connector

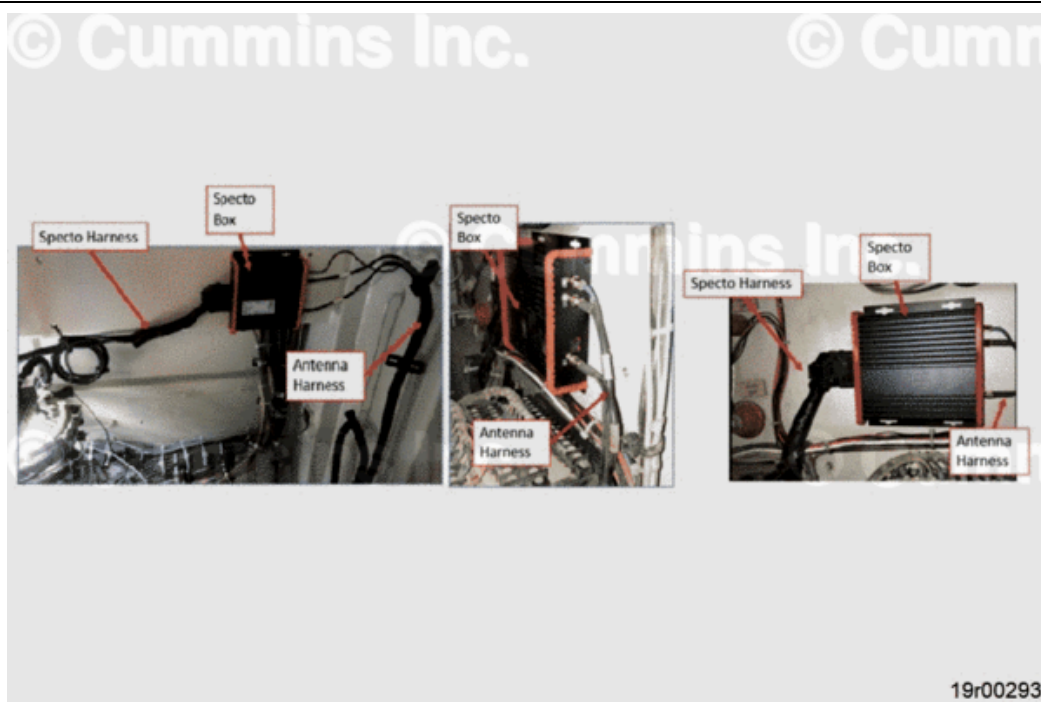


Figure 14, Telematics Box Installation Example

Appendix C: WIFI Transmission

Depending on site location, units may have either a telematics box with an external ethernet port of an internal ethernet port.

If it is determined that telematics box has an internal ethernet port, a small hole will need to be drilled above connection labeled “B” for it to be exposed.

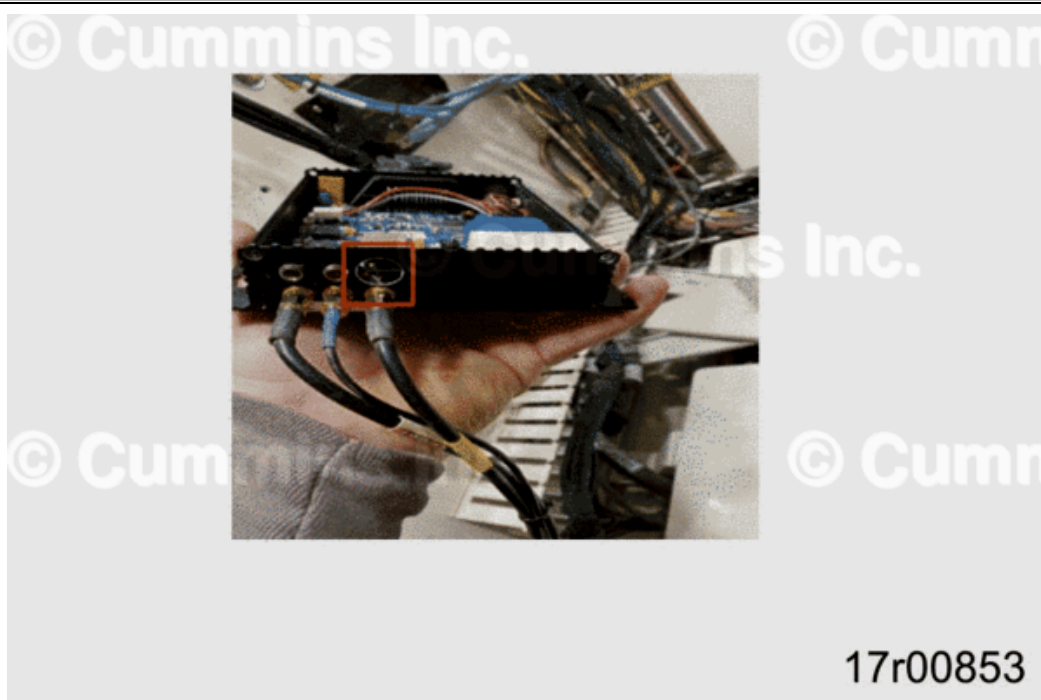


Figure 15, PrevenTech™ Telematics Box with Internal Ethernet Port

Firmware needs to be updated for both PrevenTech™ telematics box types once device can communicate online.

EMD Configuration Utility:

- EMD is a downloadable configuration program needs to configure mesh network to telematics device.
- Follow the steps below to download the program:
 1. Go to <https://www.emd.io/downloads/> (<https://www.emd.io/downloads/>)
 2. Click on second prompt called: (Windows Portable, 32bit)
 3. Use WinZip and open the program.
 4. Password: emd

Connect computer to device using a USB to RS-232 modem adapter and available RS-232 connection on device harness.



Figure 16, Device Harness RS-232 Connection

Launch EMD configuration utility and wait until dialog box appears asking for a serial port to use for communication with device, by default device will communicate at 115200 bps. Select appropriate and click OK button.

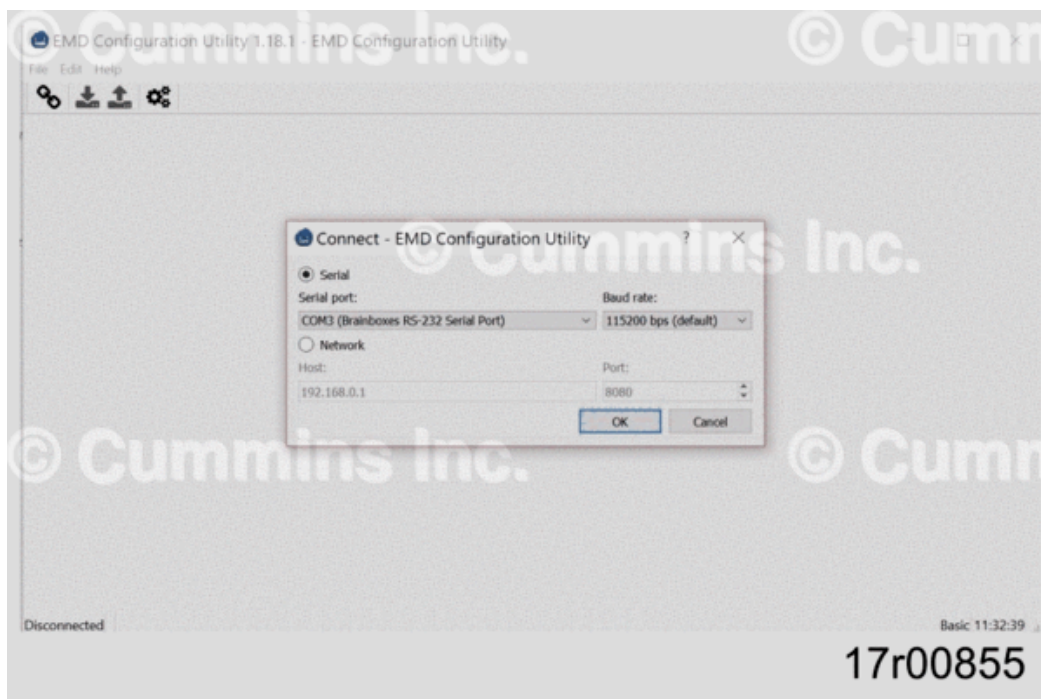


Figure 17, EMD Configuration Utility Window

A window pops up asking for a device password. Input correct password and click OK.

Note : Password of the device **must** be obtained by contacting PrevenTech™ team.



Figure 18, Device Password Input Window

DNS Server information required from customer's IT departed to link devices.

- Mode: Switch to Manual
- IP Address:
- Network Mask:
- Gateway:
- DNS1:
- DNS2:

Under the Network tab, input information gathered in appropriate locations and click Apply.

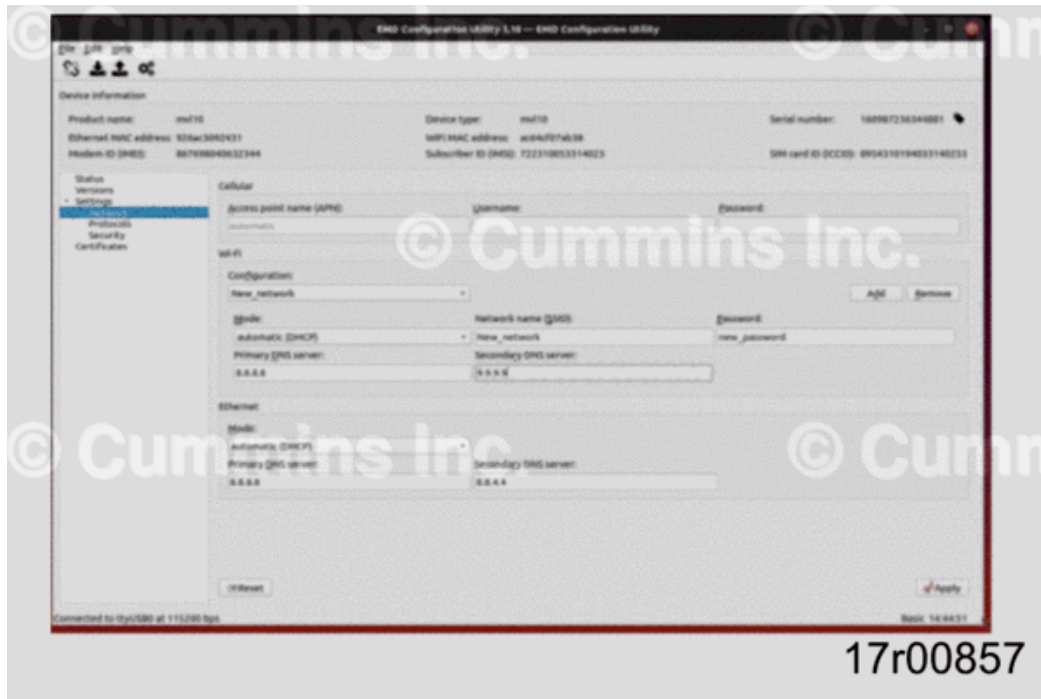


Figure 19, Network Configuration Window

Installation Considerations:

- Depending on the mine site, if the unit's current POE switch box does not have any available Ethernet ports, a new POE switch box may be needed to include the mesh network integration.
- A RJ45 – RJ45 CAT6 Ethernet cable runs from the device to the POE switch box.
 - Cable lengths vary based on location of the devices on the unit.
- A M12 (8-pin) – RJ45 CAT6 will be run from the POE switch box to the mesh network modem.
 - Cable lengths vary based on location of the devices on the unit.



Figure 20, POE/Modem Installation Example

Document History

Date	Details
2018-5-4	Module Created
2018-5-18	Minor fixes
2018-6-21	Added APPENDIX J: Troubleshooting
2018-7-13	Several Link Corrections/Troubleshooting Step Corrections
2018-8-3	none
2019-1-14	Cummins Filtration Document Link Change
2019-4-29	Updating branding on this document for PrevenTech. Also, some small updates
2020-10-12	WiFi Information Update/ Minor Fixes

Last Modified: 16-Dec-2020